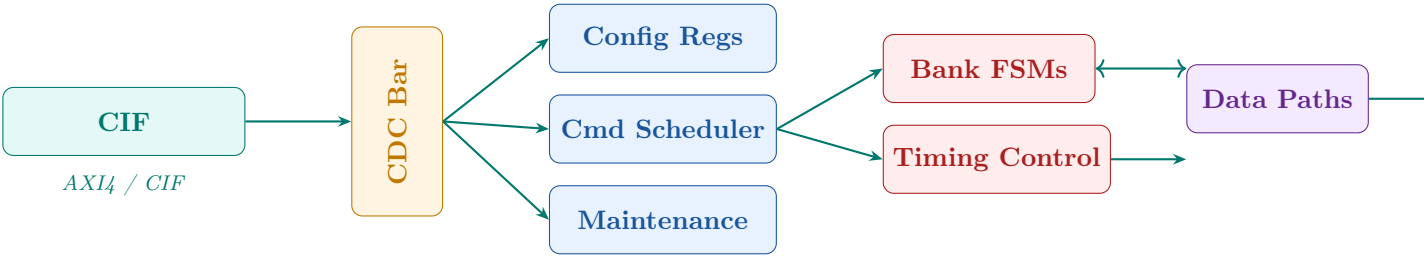


Reconfigurable DDR5 Memory Controller

MC Core — Technical Specification

Sub-Blocks · FSMs · Timing Constraints · DFI Interface



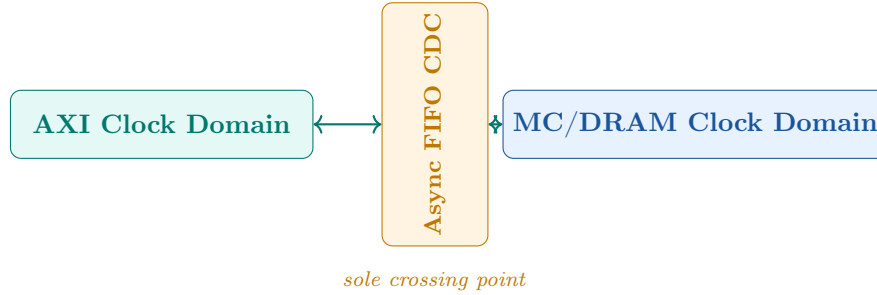
Standard	JEDEC JESD79-5 (DDR5), DFI 5.2, AXI4
Banks	16 (2 BG × 4 banks × 2 sub-channels)
Burst	BL16 (BL/2 = 8 clock cycles)
Clock	Dual domain: AXI (CIF) / MC (DRAM)
Total FSMs	≈23 (1 Init + 16 BSM + 6 Maintenance/Scheduler)

Contents

Overview and Architecture

The MC Core sits between the Client Interface (CIF) clock domain and the DRAM domain, connected via a single Async FIFO CDC bar. It is responsible for all DRAM command scheduling, timing enforcement, refresh management, power state control, and initialisation sequencing.

Clock Domains



Sub-Block Inventory

Sub-Block	Domain	Function
Config Registers	MC	Holds all timing parameters (tRCD, tRAS, tRP, tRFC, tREFI, CL, CWL, BL, ...), refresh mode, page policy, RFM thresholds. Written via AXI4-Lite CSR path.
Init FSM	MC	Full DDR5 power-up and reset sequence per JESD79-5 §3.3.1/3.3.2. Issues RESET_n, NOP burst, DLL reset, ZQcal, MRW sequence. Single instance.
Bank State Machines	MC	One FSM per bank (16 total). Tracks state (Idle/Active/Reading/Writing/Precharging/Refreshing/SR/PD) and per-bank timers.
Timing Control	MC	Per-bank timer array + cross-bank constraint checker. Computes cmd_ok[bank][cmd_type] every cycle fed to Scheduler.
DDR Cmd Scheduler	MC	Arbitrates between read/write pools, issues final DFI command. Enforces page policy (open/closed/adaptive).
Refresh Scheduler	MC	Tracks tREFI budget (leaky bucket), decides REFab vs REFsb, handles postponement up to 8×tREFI.
ZQcal FSM	MC	Periodic impedance calibration via MPC ZQCAL Start/Latch sequence.
RFM FSM	MC	DDR5 rowhammer mitigation. Per-bank RAA counter; triggers RFM command when RAA ≤ RAAIMT.
Power Management FSM	MC	CKE-controlled transitions: Active PD, Precharge PD, Self-Refresh, MPSM.
Write Data Path	MC	Write data buffer, DM/ECC generator, Write CRC computation, DFI write timing insertion.
Read Data Path	MC	Read data capture FIFO, Read CRC/ECC checker, Read latency counter, ECS FSM.

Config Registers

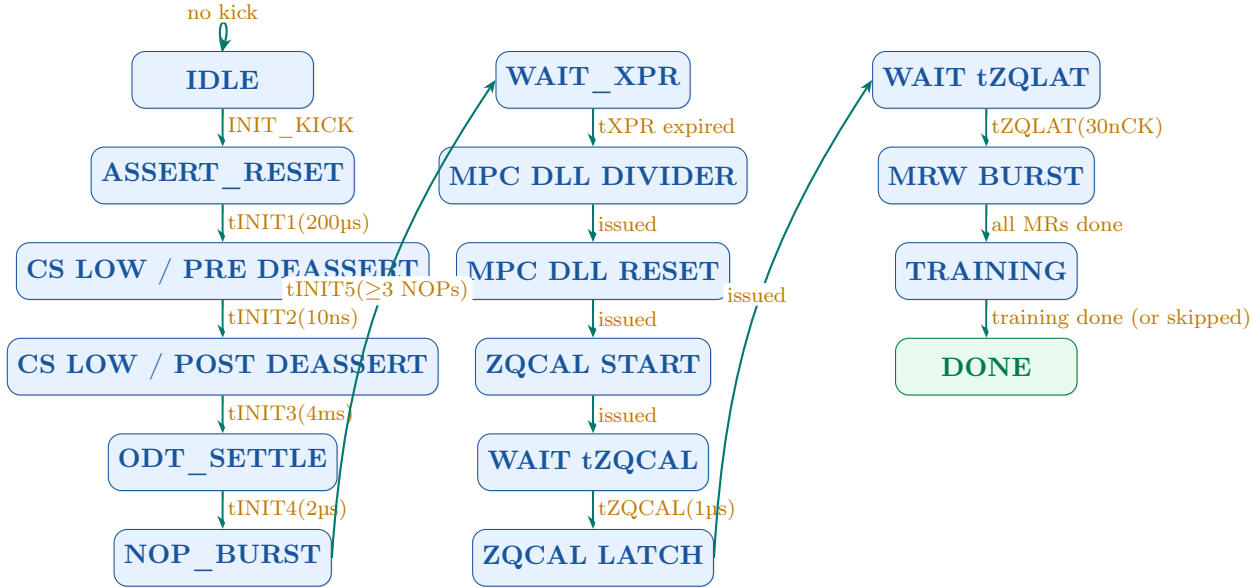
All timing knobs consumed by the MC Core are held here, programmed by the host before init kick and updatable (with care) at runtime.

Register	Width	Description
CL	7b	CAS Latency (nCK)
CWL	7b	CAS Write Latency = CL-2 (nCK)
tRCD	8b	RAS-to-CAS delay (nCK)
tRP	8b	Row Precharge Time (nCK)
tRAS	9b	Row Active Time min (nCK)
tRC	9b	Row Cycle Time = tRAS+tRP (nCK)
tWR	8b	Write Recovery Time (nCK)
tRTP	6b	Internal Read-to-Precharge (nCK)
tCCD_L	5b	CAS-to-CAS same BG (nCK)
tCCD_S	5b	CAS-to-CAS diff BG = BL/2 (nCK)
tWTR_L	6b	Write-to-Read same BG (nCK)
tWTR_S	5b	Write-to-Read diff BG (nCK)
tRRD_L	5b	ACT-to-ACT same BG (nCK)
tRRD_S	4b	ACT-to-ACT diff BG (nCK)
tFAW	6b	Four-Activate Window (nCK)
tRFC1	12b	Refresh Cycle Time 1x (nCK)
tRFCsb	11b	Same-Bank RFC (nCK)
tREFI	14b	Average Refresh Interval (nCK)
tMRD	5b	MRS-to-MRS (nCK)
tXP	5b	PDX-to-Command (nCK)
tXS	12b	SRX-to-Command no DLL (nCK)
tCKSRE	5b	Clock stable after SRE (nCK)
tCKSRX	5b	Clock stable before SRX (nCK)
tDLLK	11b	DLL Lock Time (nCK)
RAAIMT	8b	RFM RAA Initial Management Threshold
RAAMMT	8b	RFM RAA Maximum Management Threshold
RAADec	4b	RAA decrement per REF command
REF_MODE	2b	00=REFab, 01=REFsb, 10=FGR-2x, 11=FGR-4x
PAGE_POLICY	2b	00=Open, 01=Closed, 10=Adaptive
INIT_KICK	1b	Write 1 to start Init FSM
SOFT_RESET	1b	Synchronous soft-reset all FSMs

Init FSM

Reference: JESD79-5 §3.3.1 (power-up with stable power) and §3.3.2 (reset with stable power). Single instance. Triggered by **INIT_KICK**. Releases **init_done** to the Cmd Scheduler.

State Diagram



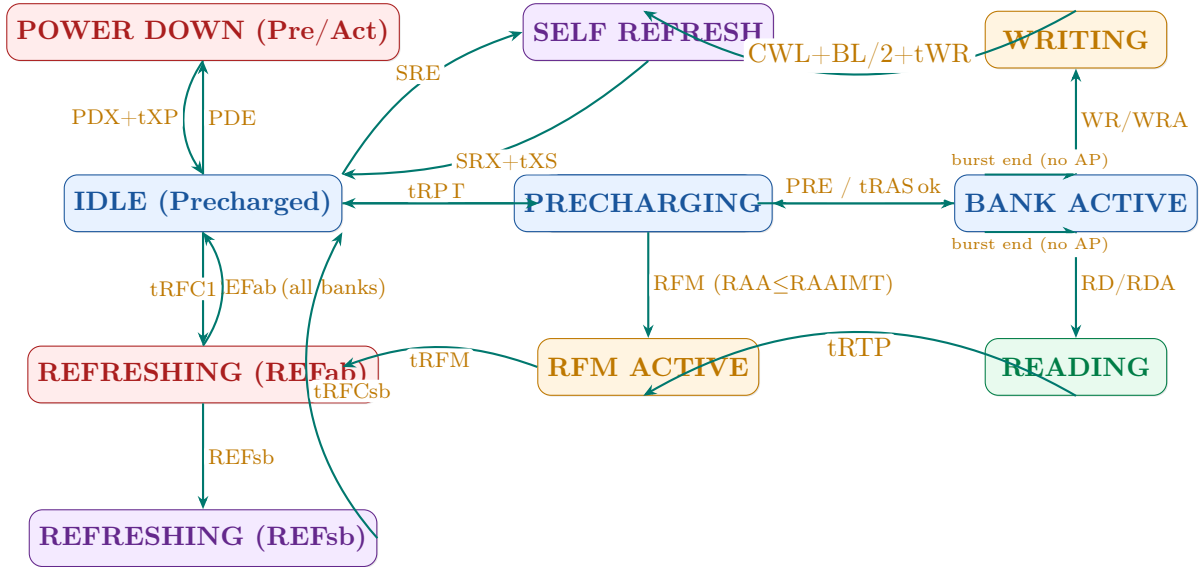
State Descriptions

State	Action / Exit Condition
IDLE	Wait for INIT_KICK =1. Assert RESET_n =0, CKE =0, CS_n =0.
ASSERT_RESET	Hold RESET_n LOW. Count 200 µs (tINIT1). CS_n held LOW. Exit when timer expires.
CS_LOW_PRE_DEASSERT	Hold CS_n LOW for $\geq t_{INIT2}=10$ ns before deasserting RESET_n .
CS_LOW_POST_DEASSERT	Deassert RESET_n (HIGH). CS_n remains LOW for tINIT3 =4 ms. CKE still LOW.
ODT_SETTLE	Raise CKE (stable CK already running). Wait tINIT4 =2 µs for ODT and CK settle.
NOP_BURST	Issue ≥ 3 NOP/DES commands (tINIT5). CS_n now driven normally.
WAIT_XPR	Wait tXPR = max(5nCK, tRFC1 +10 ns) after CS_n first high.
MPC_DLL_DIVIDER	Issue MPC: DLL Divider Select (configures tCCD_L divider via MR13).
MPC_DLL_RESET	Issue MPC: DLL Reset. Required before ZQcal per spec.
ZQCAL_START	Issue MPC: ZQCAL Start. Starts impedance calibration.
WAIT_tZQCAL	Count tZQCAL =1 µs.
ZQCAL_LATCH	Issue MPC: ZQCAL Latch. Latches calibrated driver/ODT codes.
WAIT_tZQLAT	Count tZQLAT =30 nCK.
MRW_BURST	Issue MRW to all required mode registers (MR0 , MR2 , MR4 , MR5 , MR6 , MR8 , MR13 , ...). One MRW per tMRD (16 nCK).
TRAINING_DISPATCH	Optional: CA training, CS training, Write Levelling, Read training. Controlled by TRAIN_EN bits.
DONE	Assert init_done . Release Cmd Scheduler from reset.

Bank State Machines (BSM)

16 instances (one per bank). Each BSM receives commands from the Cmd Scheduler and `cmd_ok` enable from Timing Control. All bank-private timers live here.

State Diagram (single bank)



Bank State Descriptions

State	Legal Input Commands	Timer / Exit Condition
IDLE	ACT, REFAb, REFSb, PDE, SRE, RFM	No timer. Exit on received command.
ACTIVATING	(none, internal)	tRCD counter. Transition to ACTIVE when expired.
BANK ACTIVE	RD, RDA, WR, WRA, PRE, PREA	tRAS timer ensures minimum row open time. PRE only after tRAS satisfied.
READING	BST (early termination)	CL + BL/2 until burst end. If RDA: AP started internally at tRTP.
WRITING	BST (early termination)	CWL + BL/2 until burst end. If WRA: AP started at CWL+BL/2.
PRECHARGING	(none)	tRP counter. Transition to IDLE when expired.
REFRESHING (REFab)	(none)	tRFC1 counter. All 16 banks forced into this state simultaneously.
REFRESHING (REFsb)	(none)	tRFCsb counter. Only target bank locked; others remain accessible.
POWER DOWN	PDX	Precharge PD (from IDLE) or Active PD (from ACTIVE). tCKE minimum low time.
SELF REFRESH	SRX	tCKSRE after entry; tCKESR minimum SR time; SRX raises CKE; tXS before commands.
RFM ACTIVE	(none)	tRFM counter. Issued by Maintenance Engine when $RAA \leq RAAIMT$.

Per-Bank Timer Summary

Timer	Loaded By	Condition Checked Before
tRCD counter	ACT issued	Enabling RD/WR on same bank
tRAS counter	ACT issued	Enabling PRE/PREA on same bank
tRC counter	ACT issued	Enabling next ACT on same bank
tRP counter	PRE issued	Enabling ACT on same bank
tWR counter	WR burst end	Enabling PRE on same bank
tRTP counter	RD burst start	Enabling PRE on same bank
tRFC1 counter	REFab issued	Enabling any cmd after refresh
tRFCsb counter	REFsb issued	Enabling cmd on refreshed bank
tXS counter	SRX asserted	Enabling ACT after self-refresh exit
tXP counter	PDX asserted	Enabling cmd after power-down exit
tMRD counter	MRW issued	Enabling next MRW

Timing Control

The Timing Control block is not an FSM — it is a *combinational + registered timer array* that outputs `cmd_ok[15:0]` [CMD_TYPES] every cycle.

Cross-Bank Constraints

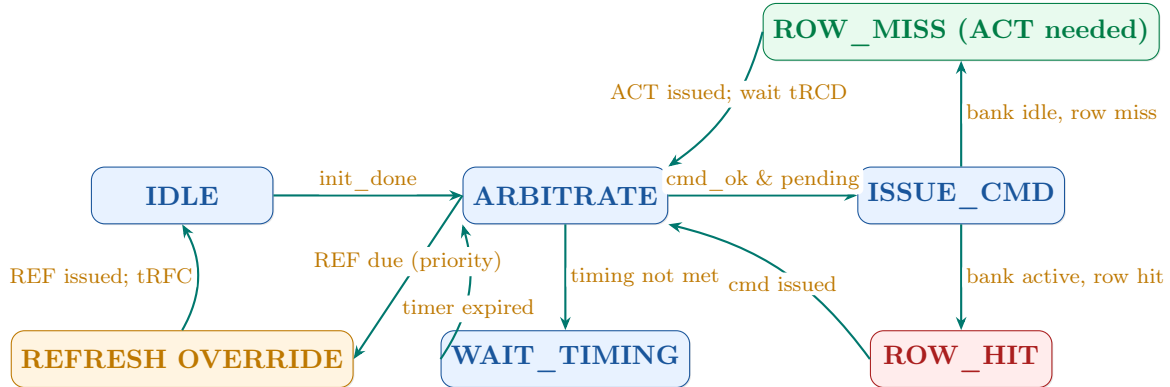
Constraint	Scope	Implementation
tRRD_L	Same BG, any bank	Timestamp of last ACT in same BG; check $T_{\text{now}} - T_{\text{last_ACT_BG}} \geq \text{tRRD_L}$
tRRD_S	Diff BG, any bank	Timestamp of last ACT globally; check $T_{\text{now}} - T_{\text{last_ACT_any}} \geq \text{tRRD_S}$
tFAW	All banks	Sliding window: ring buffer of 4 most recent ACT timestamps. Block new ACT if oldest is within tFAW.
tCCD_L	Same BG, RD→RD or WR→WR	Timestamp of last RD/WR in same BG; enforce $\geq \text{tCCD_L}$
tCCD_S	Diff BG	Timestamp of last RD/WR globally; enforce $\geq \text{tCCD_S}$
tWTR_L	Same BG, WR→RD	Timestamp of last write data end in same BG; enforce $\geq \text{WL} + \text{BL}/2 + \text{tWTR_L}$
tWTR_S	Diff BG, WR→RD	Same but tWTR_S window; enforce $\geq \text{WL} + \text{BL}/2 + \text{tWTR_S}$
tRTW	Any, RD→WR	Timestamp of last RD; enforce $\geq \text{RL} + \text{BL}/2 - \text{WL} + 2 + \text{tWPRE}$

DDR Command Scheduler

The Scheduler is the arbitration heart of the MC Core. It consumes `cmd_ok` from Timing Control, bank states from BSMs, and pending transactions from the Read/Write Cmd Pools. It issues

one DFI command per clock cycle.

Scheduler FSM



Page Policy

Policy	Behaviour
Open Page	Row stays open after RD/WR. Next access to same row = row hit (no ACT needed). Best for sequential/localised access.
Closed Page	Auto-precharge every RD/WR (RDA/WRA). Row is always idle. Best for random access patterns.
Adaptive	Track hit/miss ratio per bank. Switch to closed-page if miss rate exceeds threshold (configurable in PAGE_POLICY_THRESH).

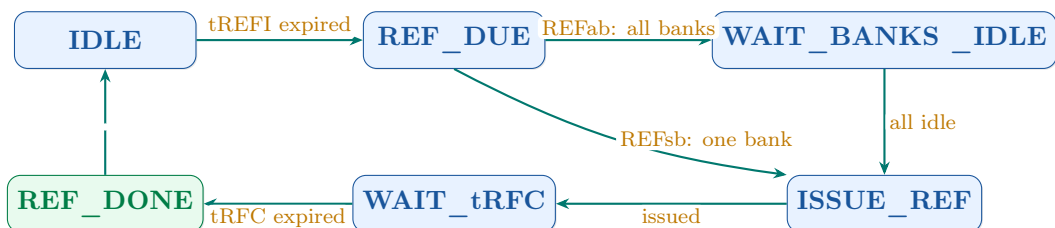
Command Priority Order

1. **Refresh Override** (REF budget about to expire — highest priority)
2. **RFM** ($RAA \leq RAA_{MT}$ — safety critical)
3. **ZQcal** (periodic, low urgency until overdue)
4. **Write drain** (if write buffer nearly full)
5. **Row-hit reads** (open page, same row)
6. **Row-hit writes**
7. **Row-miss reads** (ACT → wait tRCD → RD)
8. **Row-miss writes**
9. **PRE** (close idle banks if closed-page policy)

Maintenance Engine

The Maintenance Engine houses four sub-FSMs that handle all autonomous background operations. Each sub-FSM runs independently and posts requests to the Scheduler via a priority request bus.

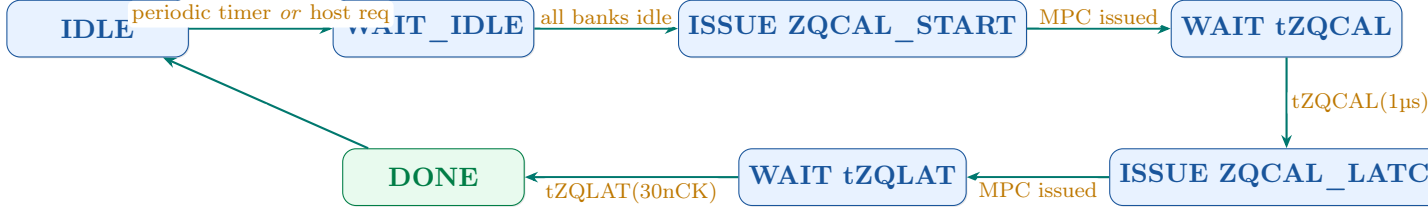
6a — Refresh Scheduler FSM



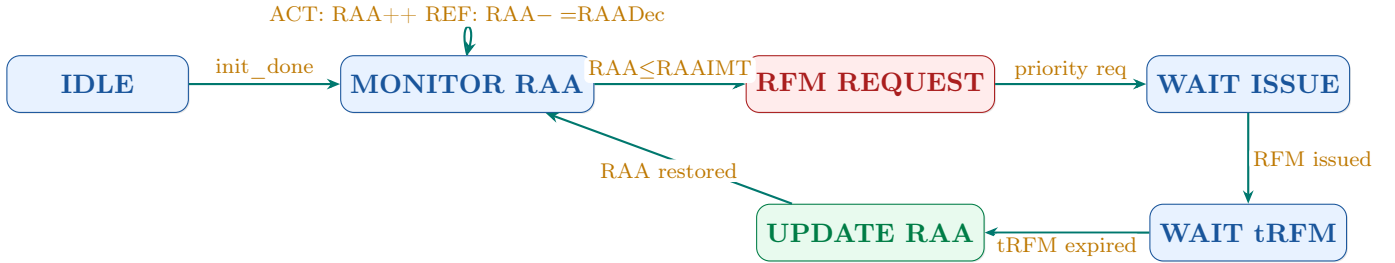
Leaky-bucket logic: A counter increments by 1 every tREFI. If it reaches 9 (max postponement = $8 \times \text{tREFI}$), the Refresh FSM asserts highest-priority override to the Scheduler. Decrement by 1 on each REF issued.

FGR (Fine Granularity): When `REF_MODE[1:0] = 10` or `11`, tRFC is replaced by tRFC2 or tRFC4 from Config Regs, and the budget counter threshold is halved or quartered accordingly.

6b — ZQcal FSM



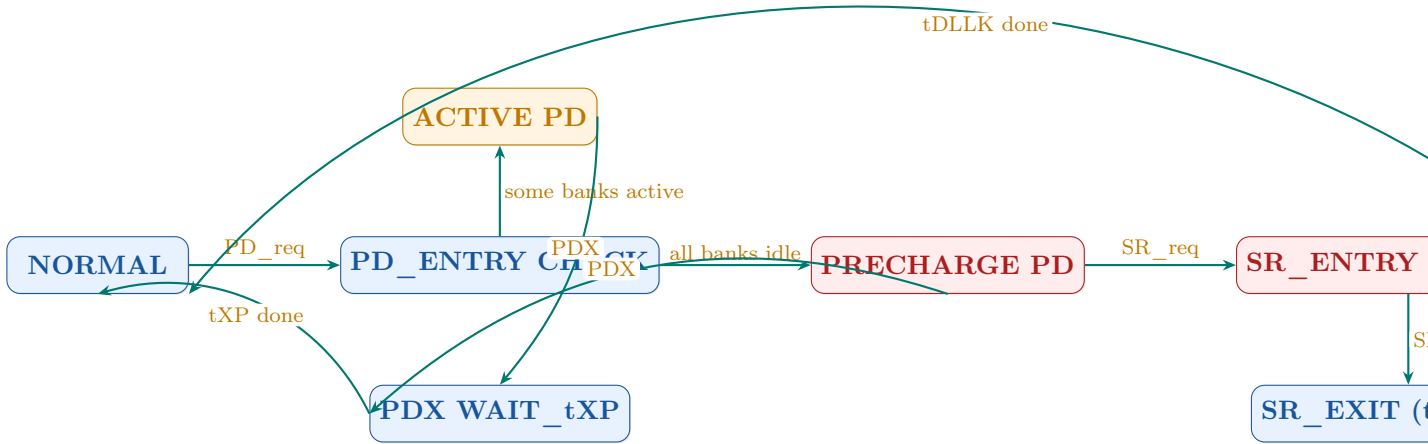
6c — RFM FSM (DDR5 Rowhammer)



RAA Logic: $\text{RAA}[b] += 1$ on every ACT to bank b ; $\text{RAA}[b] -= \text{RAADec}$ on every REF to bank b .

When $\text{RAA}[b] \leq \text{RAAIMT}$: assert RFM request (priority above ZQcal, below refresh override).

6d — Power Management FSM



Write Data Path

Sub-Block	Description
Write Data Buffer	Burst-aligned (BL16) FIFO storing data from CIF Write Cmd Pool. Dequeues when DFI write slot is confirmed by Scheduler.
DM/ECC Generator	Computes per-byte Data Mask and inline ECC. Generates Write CRC (DDR5 §4.37) using CRC-8 polynomial over DQ+DM bits.
WL Align	Inserts programmable pipeline delay = CWL to align DQS preamble to DFI <code>dfi_wrdata_en</code> .
DFI Write Timing	Drives <code>dfi_wrdata</code> , <code>dfi_wrdata_mask</code> , <code>dfi_wrdata_cs_n</code> , <code>dfi_parity_in</code> . Preamble per MR8 (2tCK default). ODT timing via <code>tODTon</code> / <code>tODToff</code> .
Write CRC Error FSM	States: <code>NORMAL</code> → <code>CRC_ERROR</code> → <code>RETRY_WR</code> → <code>AUTO_DISABLE</code> . Error detected via <code>dfi_alert_n</code> . Retry up to n times (MR51/52); auto-disable CRC if threshold exceeded.

Read Data Path

Sub-Block	Description
Read Latency Counter	Per-outstanding-read counter. Tracks CL + read preamble cycles from RD command to expected data arrival from DFI.
Read Data FIFO	Captures <code>dfi_rddata</code> when <code>dfi_rddata_valid</code> asserted. Re-aligns burst-per-read. Feeds CIF Reorder Buffer.
ECC/CRC Checker	Verifies Read CRC (§4.37.7). On CRC error: flags to CIF error register; optionally triggers read retry.

Sub-Block	Description
ECS FSM	States: IDLE → ECS_ACTIVE → WAIT_tECS → IDLE . Triggered by MR14 (auto-ECS) or host MPC request. DRAM scrubs ECC during tECS window ($\leq t_{RFC1}$).

DFI 5.2 Interface Summary

Signal	Dir	Description
dfi_address[13:0]	MC→PHY	CA bus (2-cycle for DDR5 ACT)
dfi_cs_n[1:0]	MC→PHY	Chip select (2 ranks)
dfi_cke[1:0]	MC→PHY	Clock Enable per rank
dfi_odt[1:0]	MC→PHY	On-Die Termination control
dfi_reset_n	MC→PHY	DRAM RESET_n
dfi_wrddata[127:0]	MC→PHY	Write data (BL16×DQ8)
dfi_wrddata_mask[15:0]	MC→PHY	Write data mask per byte
dfi_wrddata_en	MC→PHY	Write data enable strobe
dfi_rddata[127:0]	PHY→MC	Captured read data
dfi_rddata_valid	PHY→MC	Read data valid flag
dfi_rddata_dnv	PHY→MC	Data-not-valid (error)
dfi_alert_n	PHY→MC	DRAM alert (CRC error, parity)
dfi_ctrlupd_req	MC→PHY	Controller requests PHY update
dfi_ctrlupd_ack	PHY→MC	PHY acknowledges update
dfi_phyupd_req	PHY→MC	PHY requests MC pause
dfi_phyupd_ack	MC→PHY	MC acknowledges pause
dfi_parity_in	MC→PHY	CA parity (DDR5)
dfi_wrddata_cs_n	MC→PHY	Write CS_n (DDR5 per-DRAM)

DFI timing: `dfi_wrddata_en` must be asserted `tphy_wrlat` cycles before write data. `dfi_rddata_valid` arrives `tphy_rdlat` cycles after the RD command is issued. Both `tphy_wrlat` and `tphy_rdlat` are PHY-specific and communicated to the MC via DFI training.

FSM Summary Table

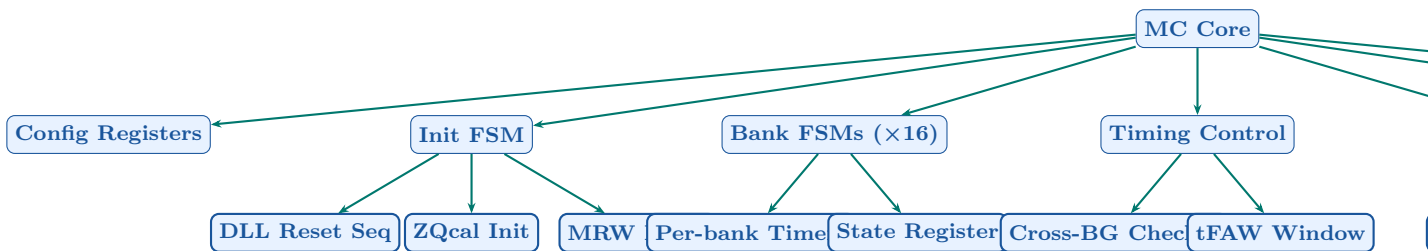
FSM	States	Instances	Notes
Init FSM	16	1	Longest; spans RESET_n to training complete
Bank State Machine (BSM)	11	16	One per bank; $16 \times 11 = 176$ state bits
DDR Cmd Scheduler	6	1	Arbitrates all command types
Refresh Scheduler	6	1	REFab/REFsb + FGR modes
ZQcal FSM	6	1	MPC ZQCAL Start/Latch flow
RFM FSM	5	1	Per-bank RAA tracking
Power Management FSM	10	1	PD (Pre/Act) + SR + MPSM
Write CRC Error FSM	4	1	Retry/auto-disable on <code>dfi_alert_n</code>
ECS FSM	4	1	DDR5 inline ECC scrub window
Total	≈68	23	

Timing Reference: Key Command Delays

All values at @200 MHz (tCK=5 ns, CL=3, CWL=1, BL/2=8) and @100 MHz (tCK=10 ns).

Command Sequence	Formula	@200 MHz	@100 MHz
ACT → RD/WR (same bank)	tRCD	4	4
ACT → PRE (same bank, min)	tRAS	7	4
PRE → ACT (same bank)	tRP	4	4
ACT → ACT (same bank)	tRC = tRAS+tRP	11	8
ACT → ACT (same BG)	tRRD_L	4	4
ACT → ACT (diff BG)	tRRD_S	2	2
RD → RD (same BG)	tCCD_L	8	8
RD → RD (diff BG)	tCCD_S = BL/2	8	8
WR → WR (same BG)	tCCD_L	8	8
WR → WR (diff BG)	tCCD_S = BL/2	8	8
RD → WR (any)	CL-CWL+BL/2+2+tWPRE	14	14
WR → RD (same BG)	CWL+BL/2+tWTR_L	13	13
WR → RD (diff BG)	CWL+BL/2+tWTR_S	11	11
RD → PRE (same bank)	tRTP	4	4
WR → PRE (same bank)	CWL+BL/2+tWR	15	12
WRA → ACT (same bank)	CWL+BL/2+tWR+tRP	19	16
RDA → ACT (same bank)	tRTP+tRP	8	8
REF → any	tRFC1	39	20
SRX → ACT	tXS = tRFC1+10ns	41	21
PDX → any	tXP	5	5
MRW → MRW	tMRD	16	16
ZQcal start → latch	tZQCAL = 1 μs	200	100
ZQCAL latch → next cmd	tZQLAT	30	30
4 ACT window constraint	tFAW	2	2

Sub-Block Hierarchy



DDR5-Specific Additions vs DDR4

Feature	DDR4	DDR5 (MC Core Impact)
REFsb (Same-Bank Refresh)	Not supported	Refresh Scheduler adds REFsb path; per-bank tRFCsb timer added to BSM
RFM (Rowhammer Mgmt)	Not supported	Full RFM FSM required; per-bank RAA counters; RAAIMT/RAAMMT Config Regs
BL	BL8 (BL/2=4)	BL16 (BL/2=8); all timing formulae scale. RTW, WTR, WR-to-PRE all longer.
ACT command width	1 cycle	2-cycle CA encoding; Init FSM and Scheduler must drive CA on two consecutive rising edges
ZQ mechanism	ZQCL/ZQCS commands	MPC ZQCAL Start+Latch; Init FSM restructured; ZQcal FSM uses MPC wrapper
tRFC1 (8 Gb)	350 ns	195 ns; shorter refresh penalty; more scheduling headroom
tREFI	7.8 μ s	3.9 μ s; double refresh rate; Refresh FSM budget counter halved
CWL definition	Separate table	CWL = CL-2; simpler Config Reg; Scheduler RTW formula simplified
tMOD	24 nCK required	Replaced by tMRD only (16 nCK); no separate tMOD path in Scheduler
ECS	Not supported	ECS FSM added in Read Data Path; hooks into tRFC window
Write CRC	Optional (MR5)	Mandatory architecture support; dfi_alert_n monitoring; retry FSM required
PDA Enumerate	Basic PDA	Enhanced per-DRAM addressing; Init FSM extended sequence